Comparing 1-3 vs. 2-2 Egg Drop Strategies with 4 Eggs and N Floors

1. The Problem Setup

We have **4 eggs** and **N floors**. We want to find the critical floor (the lowest floor from which an egg breaks) using the fewest number of drops in the worst case.

Two natural strategies emerge:

1-3 Strategy: Use one egg to make "coarse" jumps. When it breaks, switch to the 3-egg optimal strategy on the remaining interval.

2-2 Strategy: Group floors into blocks of size F. Treat each block as a "superfloor," use the 2-egg optimal strategy on the superfloors to find the critical block, then use the 2-egg optimal strategy inside that block.

Both strategies have worst-case drop counts on the order of **N^(1/4)**. Since the exponent is the same, the **constant factor** determines which is better for large N. Let's compute these constants.

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2. Review: The 1D Optimal Strategy

From the paper, Theorem 2.1 states: with **k eggs** and **N floors**, the minimum number of drops in the worst case satisfies【4†L4-L5】

$P_1(k) \leq \left\lceil k \cdot N^{1/k} \right\rceil$.P1(k) ≤ ⌈k · N1/k⌉.

For large N, the ceiling doesn't affect the asymptotic constant, so we work with

$P_1(k) \sim k \cdot N^{1/k}$.P1(k) ∼ k · N1/k.

Thus:

With **3 eggs**: $P_1(3) \sim 3N^{1/3}$P1(3) ∼ 3N1/3.

With **2 eggs**: $P_1(2) \sim 2N^{1/2}$P1(2) ∼ 2N1/2.

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3. Strategy 1-3: One Coarse Egg, Then 3 Eggs

Procedure:

Use 1 egg to test floors at intervals of size S: drop at S, 2S, 3S, ... until it breaks.

Suppose it breaks after T jumps. The critical floor lies in an interval of length S (the gap between the last safe drop and the breaking drop).

With the remaining 3 eggs, use the optimal 3-egg strategy on that interval of size S.

Worst-case drops:

Number of coarse jumps: at most N/SN/S.

Drops inside the interval: $P_1(3) \sim 3S^{1/3}$P1(3) ∼ 3S1/3.

So the total is

$D_{1\text{-}3}(S) \approx \frac{N}{S} + 3S^{1/3}$.D1-3(S) ≈ SN + 3S1/3.

Optimize S: Minimize $f(S) = NS^{-1} + 3S^{1/3}$f(S) = NS−1 + 3S1/3.

Take the derivative:

$f'(S) = -NS^{-2} + S^{-2/3} = 0 \implies S^{4/3} = N \implies S = N^{3/4}$.f′(S) = −NS−2 + S−2/3 = 0 ⟹ S4/3 = N ⟹ S = N3/4.

Plug back in:

$D_{1\text{-}3}^{\text{opt}} = \frac{N}{N^{3/4}} + 3(N^{3/4})^{1/3} = N^{1/4} + 3N^{1/4} = 4N^{1/4}$.D1-3opt = N3/4N + 3(N3/4)1/3 = N1/4 + 3N1/4 = 4N1/4.

So the **1-3 strategy** has worst-case drops

$D_{1\text{-}3} \sim 4N^{1/4}$.D1-3 ∼ 4N1/4.

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4. Strategy 2-2: Two-Level Grouping

Procedure:

Partition the N floors into blocks of size F. There are N/FN/F blocks (superfloors).

Use the 2-egg optimal strategy on the superfloors to find which block contains the critical floor. This takes $\sim 2(N/F)^{1/2}\sim$ 2(N/F)1/2 drops.

Once the critical block is identified, use the 2-egg optimal strategy inside that block of F floors. This takes $\sim 2F^{1/2}\sim$ 2F1/2 drops.

Worst-case drops:

$D_{2\text{-}2}(F) \approx 2\left(\frac{N}{F}\right)^{1/2} + 2F^{1/2}$.D2-2(F) ≈ 2(FN)1/2 + 2F1/2.

Optimize F: Minimize $g(F) = 2N^{1/2}F^{-1/2} + 2F^{1/2}$g(F) = 2N1/2F−1/2 + 2F1/2.

Take the derivative:

$g'(F) = -N^{1/2}F^{-3/2} + F^{-1/2} = 0 \quad \Longrightarrow \quad F = N^{1/2}$.g′(F) = −N1/2F−3/2 + F−1/2 = 0 ⟹ F = N1/2.

Plug back in:

$D_{2\text{-}2}^{\text{opt}} = 2\left(\frac{N}{N^{1/2}}\right)^{1/2} + 2(N^{1/2})^{1/2} = 2N^{1/4} + 2N^{1/4} = 4N^{1/4}$.D2-2opt = 2(N1/2N)1/2 + 2(N1/2)1/2 = 2N1/4 + 2N1/4 = 4N1/4.

So the **2-2 strategy** also has worst-case drops

$D_{2\text{-}2} \sim 4N^{1/4}$.D2-2 ∼ 4N1/4.

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5. Comparison and Conclusion

Strategy	Asymptotic Worst-Case Drops
1-3	$4N^{1/4}$4N1/4
2-2	$4N^{1/4}$4N1/4

Both strategies have the same leading-order constant: 4.

This means that for **large N**, the two strategies are asymptotically equivalent in the worst case. Neither is strictly better in terms of the $N^{1/4}$N1/4 scaling.

However, there may be **second-order differences**:

The 1-3 strategy has overhead from the ceiling function and the fact that the coarse jumps might not align perfectly with the optimal 3-egg search.

The 2-2 strategy has overhead from the two-level grouping and the fact that block sizes must be integers.

For **finite N**, the better strategy depends on the specific value of N and the optimal integer choices for S (in 1-3) and F (in 2-2). You would need to compute the exact integer minima for each N to determine which is better in practice.

Bottom line: For large N, both strategies achieve the same asymptotic performance, with a constant factor of 4. The choice between them is a matter of implementation convenience or second-order optimizations for specific N.